Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	20-10-2026
Team ID	Not provided
Project Name	iMovies - Movie Ticket Booking
Maximum Marks	4 Marks

Technical Architecture:

The iMovies - Movie Ticket Booking platform is designed with a scalable 3-tier architecture that includes the presentation layer (frontend), business logic (backend), and data storage layer. The solution ensures performance, security, and ease of integration with third-party APIs (e.g., Stripe for payments).

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Movie catalogue, search/filtering, showtime and seat selection, booking history	React, Vite, JavaScript, CSS
2	Application Logic	REST API for authentication, movie data, booking creation, availability, and booking history	Node.js, Express.js
3	Authentication and Security	Password hashing, JWT-based authentication, protected booking routes	bcryptjs, jsonwebtoken, Express middleware
4	Database	Stores users, movies, and booking records	MongoDB, Mongoose

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-source frameworks	Component-based frontend and REST backend	React, Vite, Node.js, Express.js
2	Data persistence	MongoDB documents accessed through object modeling	MongoDB, Mongoose
3	API architecture	Frontend communicates with backend using HTTP API requests	REST API, JSON
4	Development and version control	Local development scripts and source repository	npm, Nodemon, Git, GitHub

References:

React.js Documentation

Node js Best Practice

JSON Web Server Referance

https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d

Architecture Flow

User → React/Vite frontend → Express REST API → authentication and booking logic → MongoDB. API responses return to the frontend for display. Booking routes require authentication; seat availability is checked for the selected show.

References

React: https://react.dev/ | Vite: https://vite.dev/ | Express: https://expressjs.com/ | MongoDB: https://www.mongodb.com/docs/ | Mongoose: https://mongoosejs.com/docs/